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Ioannis M. Mandralis



I'm a PhD Candidate at Caltech specializing in Aerospace Robotics. I work on multi-modal robotic vehicles with applications in space/terrestrial exploration and mobility.

EDUCATION

California Institute of Technology
PhD candidate, GALCIT (Aerospace)

Pasadena, USA
Sept. 2021- present

ETH Zurich

MSc, Mechanical Engineering, focus on machine learning, control, and fluid dynamics
GPA: 5.84/6 (graduated with highest distinction, nominated for ETH medal.)
Dissertation: *Reinforcement learning for the control of active swimmers in flow simulations.*

Zurich, Switzerland
Sept. 2018 – Sept. 2020

McGill University

One year, merit based scholarship to study Mechanical Engineering
GPA: 4.00/4 (top <1%)

Montreal, Canada
Sept. 2017 – Sept. 2018

École Polytechnique Fédérale de Lausanne

BSc, Mechanical Engineering
GPA: 5.68/6 (est. top <1%)

Lausanne, Switzerland
Sept. 2015 – July. 2017

International School of Lausanne

International Baccalaureate
Grade: 45/45 (top 0.30% worldwide)

Lausanne, Switzerland
Aug. 2013 – June 2015

RESEARCH EXPERIENCE

Computational Science & Engineering Lab, ETH Zurich
Research Assistant, in Prof. Petros Koumoutsakos group

Zurich, Switzerland
Feb. 2019 – May 2021

Reinforcement learning for the control of active swimmers in flow simulations

- Coupled Deep Reinforcement Learning algorithms with Fluid Simulations.
- Employed learning algorithms to obtain optimal fish swimming patterns under energy constraints.
- Employed learning algorithms for micro-swimmers navigating in unknown flow fields.

Generative Adversarial Networks for Modeling Stochastic Dynamical Systems

- Designed a Generative Adversarial Network for learning distributions of stochastic dynamical systems.
- Tested the system for predicting Alanine Dipeptide protein kinetic compared to MD simulations.
- Compared the results to molecular dynamics simulations.

Engineering Mechanics of Soft Interfaces Lab, EPFL

Research Assistant, supervised by Prof. John Kolinski

Lausanne, Switzerland

June 2018 – July 2018

- Built an experimental optics setup for two-dimensional photoelasticity.
- Extended two-dimensional setup for tomography of 3D materials.

TEACHING EXPERIENCE

Optimal Control and Estimation

Teaching Assistant under Prof. Richard Murray

Caltech
Winter 2023

- Graded and gave feedback on class exercises throughout the semester.

Linear Systems Theory

Teaching Assistant under Prof. Amir Rahmani (JPL)

Caltech
Fall 2023

- Graded and gave feedback on class exercises throughout the semester.

Multivariable Control Systems for Electrical Engineers

Exercise Assistant

ETHZ
Spring 2020

- Graded and gave feedback on class exercises throughout the semester.

General Mechanics and Thermodynamics for Material Engineers

Teaching & Exercise Assistant

EPFL
Spring 2017

- Conducted 15 min blackboard lectures, followed by discussion for a class of 20 students – graded mid-terms.

Analytical Mechanics for Mechanical Engineers

Teaching & Exercise Assistant

EPFL
Fall 2016

- Solved exercises on the blackboard and answered questions for a class of 30 students.

PROFESSIONAL EXPERIENCE

ANYbotics AG

Robotics Software Developer

Zurich, Switzerland
Sept. 2019 – Apr. 2020

- Developed system identification techniques for parameter estimation in robotic systems.
- Implemented the identification software in C++ and Python for in-house, robotic system ANYmal.
- Participated in general software development tasks in C++, Python, and ROS.

Nestlé, R&D, Systems Technology Center

Mechanical Engineering Intern

Orbe, Switzerland
Summer 2017

- Designed and 3D printed mechanical parts for coffee dispensing systems using SolidWorks.

PUBLICATIONS

[1] Mandralis I., Weber P., Novati G., Koumoutsakos P., “Learning swimming escape patterns under energy constraints.” **Phys. Rev. Fluids** 6, 093101, Sept 2021. <https://doi.org/10.1103/PhysRevFluids.6.093101>

[2] Gunnarson, P., Mandralis, I., Novati, G., Koumoutsakos P. & Dabiri J O. Learning efficient navigation in vortical flow fields. **Nature Communications** 12,7143 (2021). <https://doi.org/10.1038/s41467-021-27015-y>

HONOURS & AWARDS

◊ **Onassis Fellowship**

Awarded by the Onassis Foundation to support PhD studies for a period of 5 years.

◊ **Rolf D. Buhler Memorial Prize in Aeronautics**

Awarded by the Caltech Aerospace department to student with highest GPA in first year of PhD studies.

◊ **ETH Ambassador for the Global Young Scientist Summit, Singapore, 2021**

Awarded to 10 members of ETHZ selected from entire body of Post-Docs, PhDs, and MSc Students

◊ **Scholarship for Fully Funded Exchange Year at McGill University**

Awarded by EPFL based on academic merit

SKILLS & INTERESTS

Programming: C++, Python, Matlab, ROS, PyTorch, Tensorflow, LaTeX.

Design and Simulation: SolidWorks, Catia, Abaqus, Ansys, Simulink, LabView.

Languages: English (native), Greek (native), French (fluent), Spanish (C1), German (A1).

Interests: Basketball, Music/stringed instruments (acoustic/electric guitar/bouzouki), Traveling, Economics.

REFERENCES

Prof. Petros Koumoutsakos

*Gordon McKay Professor of Computing
in Science and Engineering,
Harvard/ETHZ
petros@seas.harvard.edu*

Prof. John O. Dabiri

*Centennial Professor of Aeronautics
and Mechanical Engineering,
Caltech
jodabiri@caltech.edu*

Prof. Demetri Psaltis

*Professor in Bioengineering and
director of the Optics Laboratory
EPFL
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